



Assessing the Ecological Influence of Invasive Plant Species in Virginia

Seun Oladipo¹, Raymond Santiago¹

¹Department of Environmental Sciences, University of Virginia, USA. ¹Department of Statistics, University of Virginia, USA.

Introduction

Background: Virginia is home to 103 invasive plant species. These species alter the functioning of infested forested ecosystems, including nutrient cycling, carbon sequestration, reproduction and net primary productivity.

Research Questions:

- Where are the invasive species dominant in Virginia?
- What are the environmental factors driving the distribution of invasive plant species?

Research Goals

- Produce a georeferenced distribution map of all invasive plant occurrences in Virginia
- Develop habitat suitability maps for focal species using Species Distribution Modeling (SDM)
- Identify dominant climatic variables driving invasive species presence using Random Forest variable importance
- Quantify distributional differences in climate niches across species using nonparametric tests
- Identify high-priority county-level invasion hotspots for management intervention

Method

01 Data Collection

- USFS GRIPHO: 36,935 records
- WorldClim 2.1 (bio01, bio04, bio12, bio15)
- VIPS 2024 species validation list

02 Data Cleaning

- Remove duplicates & set coordinates (EPSG:4326)
- Filter to Virginia invasive species
- Final dataset: 822 observations

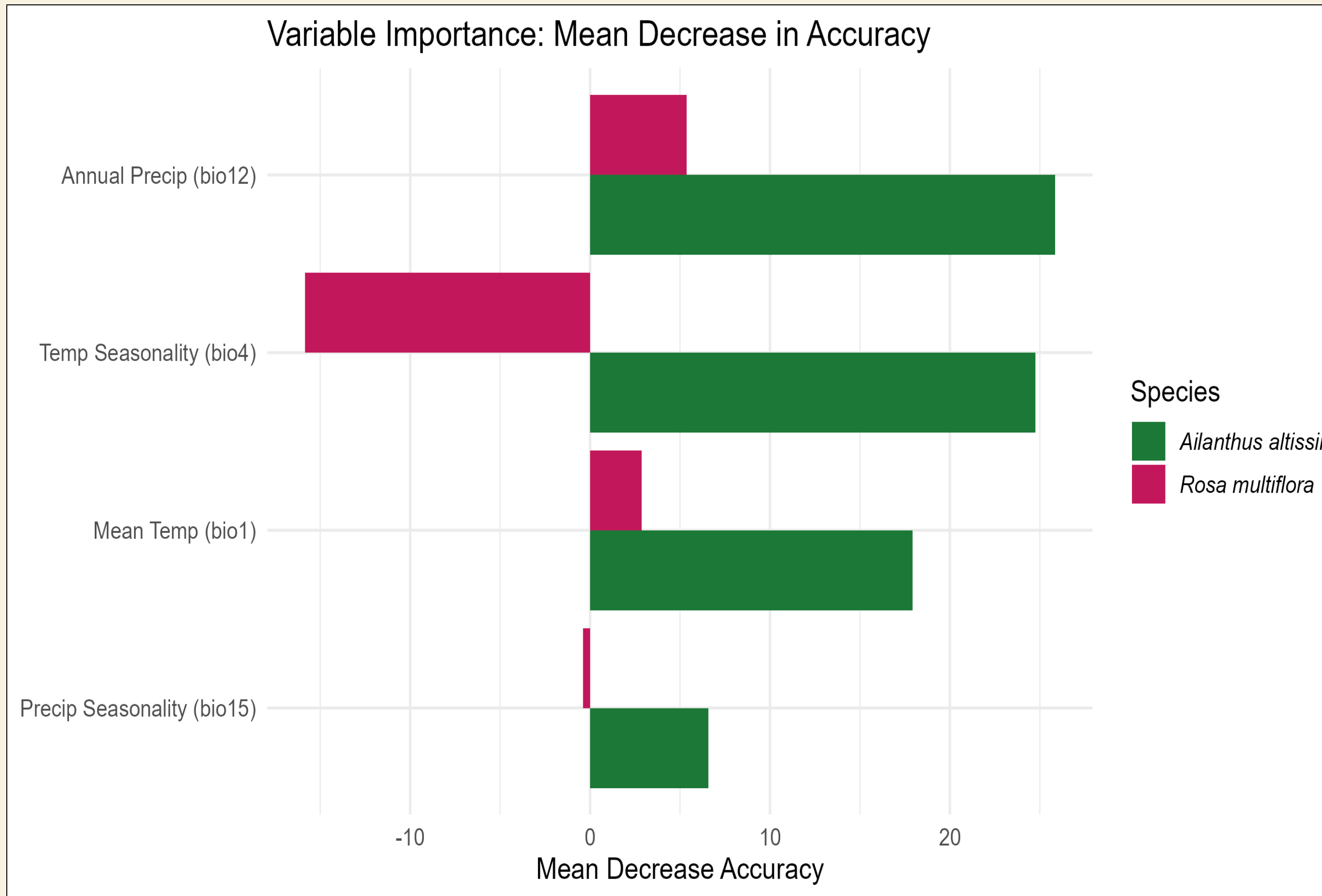
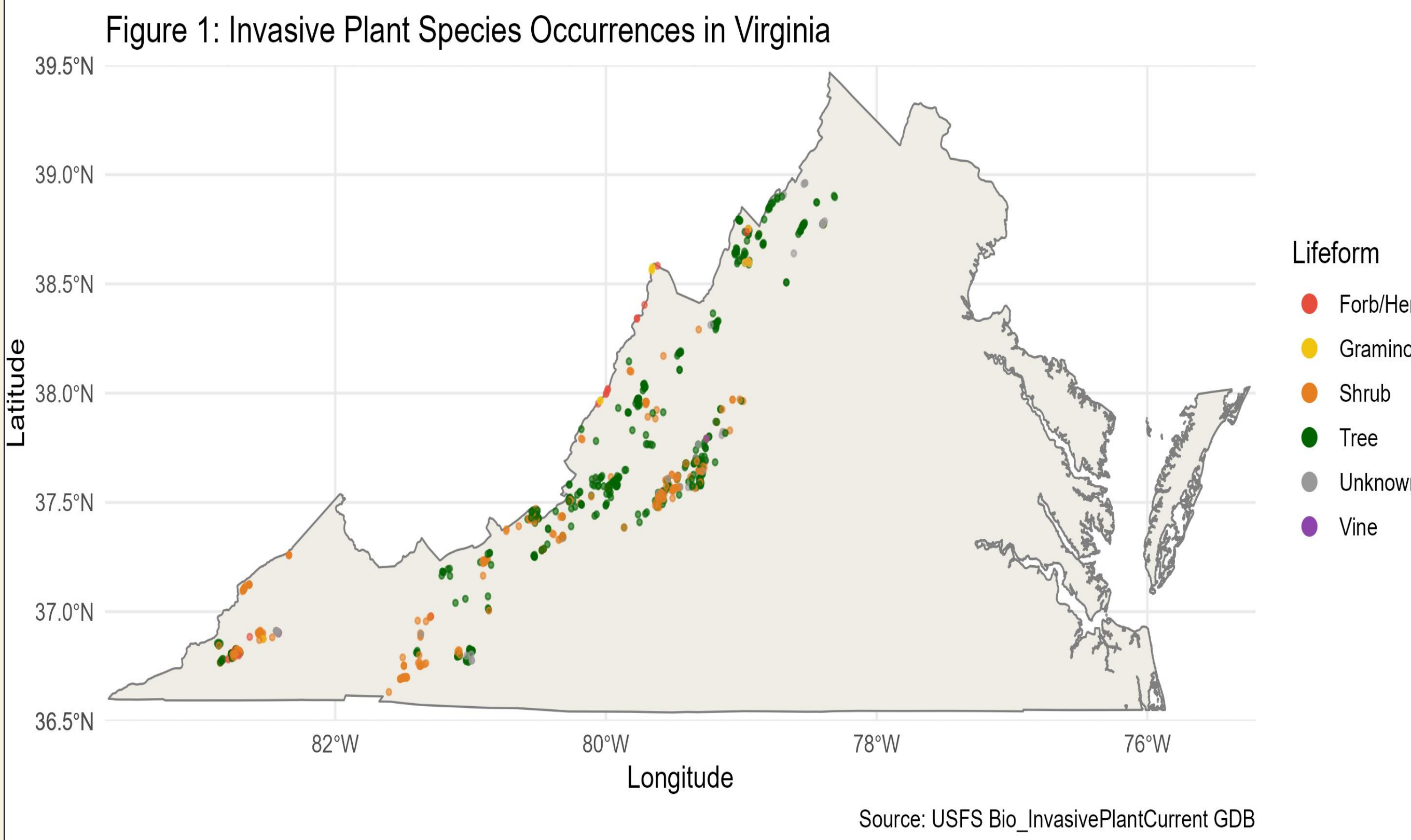
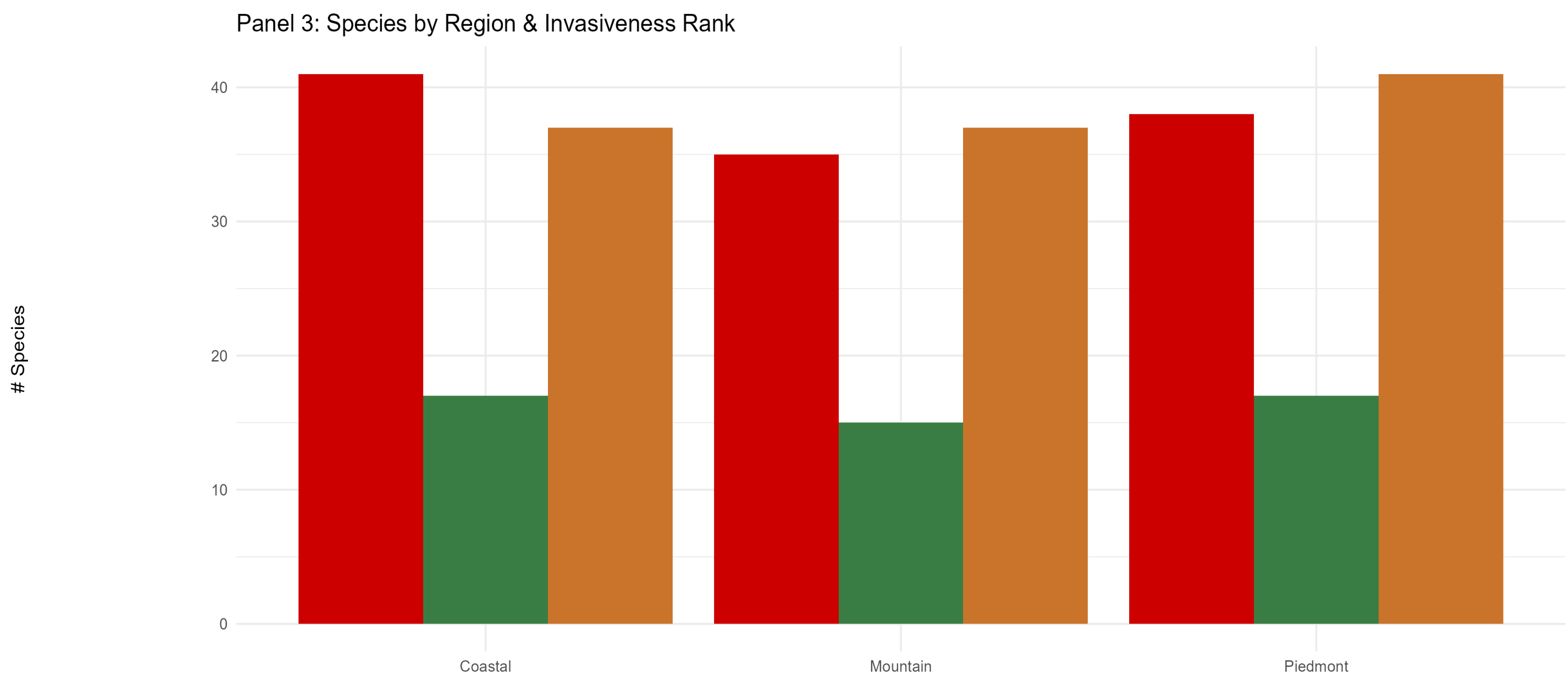
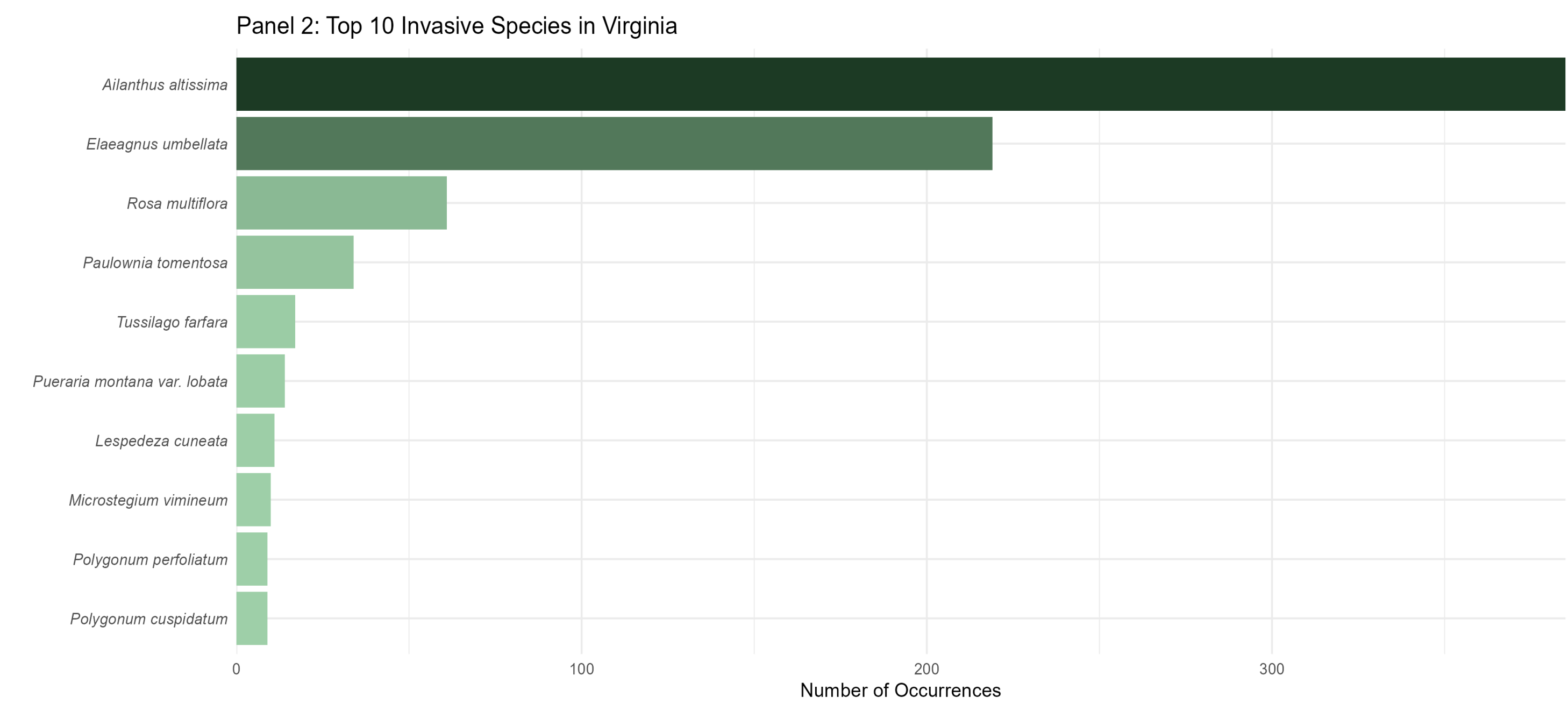
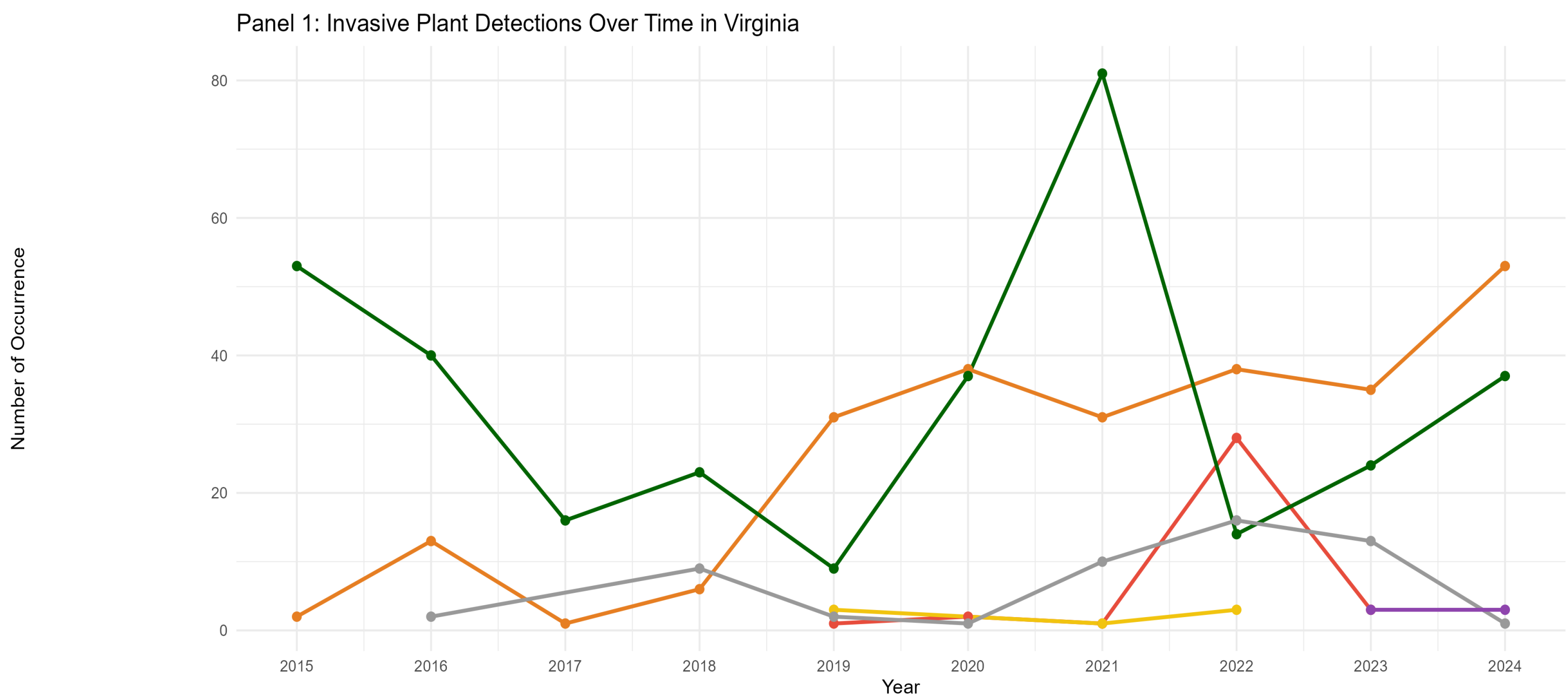
03 Analysis

- Species Distribution Modelling
- Kruskal-Wallis climate niche tests
- Hotspot Analysis

04 Outputs

- VA-wide distribution map
- Habitat suitability (focal species)
- Climate driver rankings
- County-level invasion hotspots

Results



Conclusions

The study mapped invasive plant species distribution in Virginia, which are usually found in mountains, piedmont and coastal areas. The findings revealed that most of the Invasive species are found in and around the Appalachian mountains. Of the 822 invasive plant species in Virginia, the top three counties where they are located are Botetourt (68), Amherst (67), and Rockbridge (62). The top three species are Ailanthus altissima (Tree, n=382), Elaeagnus umbellata (Shrub, n=219), Rosa multiflora (Shrub, n=63). The contributing factors are human influence, changing climate and adaptive ability of these invasive species.

References

